

C Programming 01

Search Algorithm

Pre-learning Video

- If you have not watched the pre-learning video before working on the exercises, please do so first.
 - <https://wsdmoodle.waseda.jp/mod/millvi/view.php?id=5822908>

linear search and binary search

- **Linear search** is an algorithm for searching data from an array.
- **Binary search** is an algorithm for searching data from an array. The data should be sorted.
- Write the following program using each of these two algorithms.

Exercise1

- Prepare 100 random values (1-100) and store them in an array.
- Create a program that searches for the number given from the keyboard in the prepared array.
 - If the number is found, print the location in the array and "the number exists"
 - If there is no applicable number, display "Number does not exist".

Exercise2

- Prepare the following array.
 - ["Java", "Python", "C", "Ruby", "` Basic"]
- Write a program to search "C" from this array.
 - do not use the standard library.
 - string.h and ctype.h
- Indicate at what position "C" was found in the resulting array.
 - Try both if you can.
 - Linear Search
 - Binary search

Exercise 3

Problem: Find the K-th Smallest Number

You are given an integer array `arr`, and your task is to find the K-th smallest number in the array.

Input

- The first line contains an integer N ($1 \leq N \leq 10^5$), representing the number of elements in the array.
- The second line contains N integers `arr[i]` ($1 \leq \text{arr}[i] \leq 10^9$).
- The third line contains an integer K ($1 \leq K \leq N$), representing the position of the smallest number to find.

Output

- Print the K-th smallest element as an integer.

Constraints

- Since N is large, sorting the array may be inefficient.
- Consider optimized algorithms such as QuickSelect or a Min-Heap for better performance.

Sample Input

```
6
7 10 4 3 20 15
3
```

Sample Output

```
7
```

Exercise Submission

- Report 2 will be conducted on June 15 for the 9th class.
 - It covers the contents of the 4th through 8th classes.
 - Report 2 will contain several exercises, including both practice exercises and application exercises.
 - Since we are reusing last year's Moodle, you may see links to Zoom, etc., but we will not be using them. I will make a separate announcement when I will be giving lectures online.